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Design Simulation and Analysis of Hybrid LMS-RLS Adaptive Filters Radio Wave Signal Processing in Radar

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ABSTRACT This study investigates the implementation and performance analysis of adaptive filters based on LMS (Least Mean Squares), RLS (Recursive Least Squares), and two hybrid approaches—cascade and parallel mixture—to enhance radar signals corrupted by Gaussian white noise and colored noise. Simulations are carried out in MATLAB using a chirp signal as the primary input with two noise scenarios as interference. Evaluation is performed using SNR (Signal-to-Noise Ratio) and MSE (Mean Squared Error), both numerically and through waveform visualization and adaptive mixing coefficient $\alpha(n)$. In the white noise scenario, LMS improved the SNR from −6.70 dB to −0.38 dB and reduced the MSE from 2.68723 to 1.01852. Meanwhile, in the colored noise scenario, the performance improvement is relatively small, with the SNR increasing from −10.31 dB to 0.07 dB and the MSE decreasing from 5.32817 to 1.07686. Based on these results, LMS performs reasonably well for white noise but remains less effective for colored noise due to its temporal correlation. The cascade LMS-RLS approach showed better performance compared to the standalone LMS. For white noise, cascade increased the SNR to 1.68 dB and reduced the MSE to 0.73961. In the colored noise scenario, its performance improved further, achieving an SNR of 1.08 dB and an MSE of 0.86476. These results demonstrate that the selective error-based correction mechanism in the cascade method effectively compensates for the limitations of LMS, especially in correlated noise environments. Meanwhile, the parallel mixture approach produced the best performance in all scenarios. For white noise, the parallel mixture achieved an SNR of 4.02 dB and an MSE of 0.48895, while in the colored noise scenario it achieved an SNR improvement up to 5.18 dB and reduced the MSE to 0.45637. Given its consistent superior performance in both conditions, the parallel mixture method stands out as the most optimal adaptive filtering solution for radar systems operating in complex noisy environments.

INDEX TERMS Adaptive filters, Cascade LMS-RLS, Colored noise, Hybrid filters, Least Mean Squares (LMS), Parallel mixture, Radar signal processing, Recursive Least Squares (RLS), Signal-to-Noise Ratio (SNR)

I. INTRODUCTION

RADAR systems have experienced significant advances, expanding from military applications to the civilian sectors such as automotive safety and remote sensing [1]. A critical challenge in modern radar signal processing is maintaining optimal detection performance in environments contaminated by noise and interference [1]. To enhance the detectability of echo signals, particularly in Pulse Compression radar systems utilizing Linear Frequency Modulation (LFM) waveforms, adaptive filtering techniques are essential [4].

The two most established algorithms for adaptive filtering are the Least Mean Squares (LMS) and Recursive Least Squares (RLS). LMS is renowned for its computational simplicity and ease of implementation. However, it often exhibits slow convergence rates and reduced effectiveness in correlated (colored) noise environments [2]. In contrast, RLS provides fast convergence and robustness against signal variations, but requires significantly higher computational resources [2]. Consequently, relying on a single algorithm

often requires a compromise between performance and efficiency.

To address these limitations, this study proposes Hybrid LMS-RLS architectures. By integrating the simplicity of LMS with the precision of RLS, hybrid filters aim to optimize noise reduction. This research investigates two specific hybrid configurations: a Cascade structure, which selectively activates RLS based on an error threshold to conserve computational power, and a Parallel Mixture structure, which dynamically combines the outputs of both filters using an adaptive mixing coefficient, $\alpha(n)$.

The primary contribution of this work is a comprehensive performance analysis of these hybrid filters compared to standalone LMS and RLS. Simulations are conducted using LFM chirp signals corrupted by both White Gaussian Noise and Colored Noise. The evaluation metrics include Signal-to-Noise Ratio (SNR) improvement, Mean Squared Error (MSE) reduction, and waveform visualization [7]. The results indicate that the parallel mixture method delivers superior performance, especially in complex colored noise scenarios.

The remainder of this paper is organized as follows. Section II reviews the radar signal model and adaptive filter theories. Section III details the design of the proposed hybrid structures (Cascade and Parallel). Section IV presents the simulation results and analysis, and Section V concludes the study.

II. RADAR TECHNOLOGY AND ADAPTIVE FILTERS

A. PULSED RADAR

The radar system operates by transmitting a short pulse of electromagnetic energy and subsequently switching to a receiver mode to detect the echo reflected from a target. The target's range, R, is estimated based on the round-trip time delay, τ, between the signal transmission and the reception of the echo. This relationship is defined as:

$$R = \frac{c\tau}{2} \tag{1}$$

where c represents the speed of light (3×10^8 m/s). The factor of 1/2 accounts for the two-way propagation path of the signal.

The system is further characterized by the Pulse Repetition Interval (PRI), which dictates the timing between consecutive pulses. This parameter establishes the maximum unambiguous range, R_{max}, beyond which range aliasing occurs:

$$R_{max} = \frac{c \cdot PRI}{2} \tag{2}$$

B. LINEAR FREQUENCY MODULATION (CHIRP)

To decouple range resolution from the transmitted pulse energy, the radar system employs Linear Frequency Modulation (LFM), commonly referred to as a chirp. Unlike a simple unmodulated pulse, the instantaneous frequency of an LFM waveform varies linearly with time during the pulse duration. This modulation technique enables the transmission of long-duration pulses to maximize total energy—thereby enhancing detection performance—while simultaneously achieving fine range resolution through pulse compression at the receiver.

The complex baseband representation of the transmitted LFM signal, $s_{tx}(t)$, is defined as:

$$s_{tx}(t) = A \cdot \exp\left[j2\pi\left(f_0 t + \frac{1}{2}St^2\right)\right], \quad 0 \le t \le T_p \tag{3}$$

where A denotes the signal amplitude, f_0 is the starting sweep frequency, and T_p represents the pulse width. The parameter S is the chirp slope, which describes the rate of frequency change and is related to the sweep bandwidth B by:

$$S = \frac{B}{T_p} \tag{4}$$

By controlling the bandwidth B independently of the pulse width T_p, the system achieves a time-bandwidth product much greater than unity, which is the fundamental requirement for effective pulse compression.

C. PULSE COMPRESSION

A fundamental challenge in pulsed radar systems is the inherent trade-off between detection range and range resolution. Short pulses provide superior range resolution but lack the energy required for long-range detection, whereas long pulses improve detection capability but degrade resolution. Pulse compression resolves this conflict by transmitting a long, modulated pulse (such as the LFM chirp defined previously) and compressing it at the receiver side [36], [39].

The compression is typically implemented using a matched filter, which is the optimal linear filter for maximizing the Signal-to-Noise Ratio (SNR) in the presence of additive white Gaussian noise. The impulse response of the matched filter, $h(t)$, is the time-reversed and complexconjugated version of the transmitted signal:

$$h(t) = s_{tx}^*(T_p - t) \tag{5}$$

The output of the pulse compression, $y(t)$, is obtained by convolving the received signal, $r(t)$, with the matched filter response:

$$y(t) = (r * h)(t) \tag{6}$$

This process concentrates the energy of the long pulse into a narrow peak at the target's time delay. Consequently,

the range resolution, ΔR, becomes a function of the signal bandwidth, B, rather than the pulse duration, allowing for high-resolution imaging even with high-energy pulses:

$$\Delta R \approx \frac{c}{2B} \tag{7}$$

D. RADAR SYSTEM ARCHITECTURE

The radar system in this study is modeled as a monostatic pulsed radar employing Linear Frequency Modulation (LFM). The simulation architecture encompasses the complete signal chain: waveform generation at the transmitter, propagation effects including target delay and Doppler shift, and signal processing at the receiver via pulse compression. The overall system architecture is illustrated in Fig. 1.

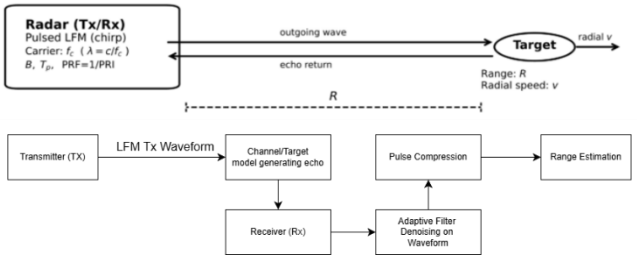


FIGURE 1. Block diagram of the monostatic pulsed LFM radar system showing the signal flow from transmission to range estimation.

Mathematically, the received baseband echo signal, $r(t)$, from a moving target is modeled as a delayed and attenuated version of the transmitted signal, superimposed with noise:

$$r(t) = \alpha s_{tx}(t - \tau)e^{j2\pi f_D t} + n(t) \tag{8}$$

where α represents the complex attenuation factor (accounting for the radar equation, RCS, and propagation losses), τ is the round-trip time delay corresponding to the target range R ($\tau = 2R/c$), and $n(t)$ denotes the additive noise component.

The term f_D represents the Doppler frequency shift induced by the radial velocity, v, of the target. For a monostatic radar configuration, this shift is approximated as:

$$f_D = \frac{2v}{\lambda} = \frac{2vf_c}{c} \tag{9}$$

where f_c is the carrier frequency and λ is the wavelength. Although the simulation is performed in the baseband domain, explicit definition of parameters such as f_c, chirp bandwidth B, and pulse width T_p is crucial to accurately model the phase relationships and Doppler effects inherent in the radar echo [36], [40].

E. NOISE

In operational environments, the received radar signal is inevitably contaminated by noise arising from thermal fluctuations, environmental clutter, or intentional interference. This degradation poses a significant challenge to target detection. To rigorously evaluate the robustness of the proposed adaptive filters, this study models two distinct noise categories: uncorrelated white noise and correlated colored noise.

1) Gaussian White Noise

The first model is Gaussian White Noise (WGN), which serves as a baseline for ideal background interference. This noise is characterized by a normal distribution with a zero mean and a constant variance, σ^2. A defining property of WGN is that its samples are statistically independent, exhibiting an impulsive autocorrelation function ($R_n(\tau) = \sigma^2\delta(\tau)$). The probabilistic model is expressed as:

$$n(t) \sim \mathrm{N}(0,\sigma^2) \tag{10}$$

In the frequency domain, WGN possesses a uniform Power Spectral Density (PSD) across the entire bandwidth ($S_n(f) = N_0/2$). While useful for fundamental analysis, relying solely on WGN often yields overly optimistic performance estimates for radar systems operating in complex environments.

2) Colored Noise

The second model is Colored Noise, employed to simulate realistic scenarios where interference exhibits temporal correlation and a non-uniform spectral distribution, such as clutter or specific jamming modulation. Unlike white noise, colored noise retains memory of previous samples.

Mathematically, the colored noise component, $n_c(t)$, is synthesized by passing white Gaussian noise, $n_w(t)$, through a linear shaping filter with an impulse response $h(t)$. This process is defined by the convolution operation:

$$n_c(t) = h(t) * n_w(t) \tag{11}$$

Consequently, the PSD of the colored noise is determined by the frequency response of the shaping filter, $H(f)$, as follows:

$$S_{nc}(f) = |H(f)|^2 S_{nw}(f) \tag{12}$$

In this simulation, an Autoregressive (AR) model is utilized to generate the correlated noise sequences, providing a rigorous testbed for evaluating the adaptive capabilities of the hybrid filter architectures.

III. METHODOLOGY

In this study, simulations are conducted to analyze the performance of various adaptive filter algorithms applied to radar signals generated from a radar system model corrupted by both white and colored noise. Unlike conventional synthetic chirp approaches, the clean signal in this study is generated using a monostatic pulsed radar model with Linear Frequency Modulation (LFM). This model simulates the complete physical radar process: transmission, propagation, target reflection, and reception (Tx-channel-target-Rx). Consequently, the signal processed by the adaptive filters represents a radar echo waveform, mathematically encapsulating range (delay) and velocity (Doppler) information. Mathematical modeling and the implementation of Least Mean Squares (LMS), Recursive Least Squares (RLS), and their hybrid combinations (cascade and parallel architectures) were developed using MATLAB. The research focuses on assessing the effectiveness of each approach in enhancing the Signal-to

Noise Ratio (SNR) and minimizing the Mean Squared Error (MSE). The simulation process encompasses two distinct noise scenarios: Gaussian white noise and colored noise modeled using an Autoregressive (AR(1)) filter. The output of each filter algorithm is analyzed based on numerical performance (SNR and MSE) and visual representations, including running MSE curves, filtered signal waveforms, and mixing coefficient (λ) graphs for the parallel model. Additionally, code structure, comprehensive pseudocode, and simulation flowcharts are utilized to provide systematic documentation of the modeling process.

A. SIMULATION PARAMETERS

The simulation is performed in the discrete-time domain with a sampling frequency of f_s = 2 MHz. The observation duration is defined as T_{obs}, resulting in a total number of samples $N = T_{obs} \cdot f_s$. The transmitted radar signal utilizes a Linear Frequency Modulation (LFM) pulse, or chirp, with a pulse duration T_p and bandwidth B, where the slope is defined as S = B/T_p.

The adaptive filter order is set to $M = 32$ for all methods, including LMS, RLS, and their hybrid combinations. The learning rate (step size) for the LMS algorithm is fixed at $\mu = 0.001$. Two distinct noise scenarios are employed: white Gaussian noise with a standard deviation of $\sigma = 1.5$, and colored noise with $\sigma = 0.5$, which is generated using an

Autoregressive model of order 1 (AR(1)) with a pole at 0.9.

For the RLS algorithm, the parameters include a forgetting factor of $\lambda_{RLS} = 0.995$ and an initial regularization parameter of $\delta_{RLS} = 1$. In the cascade LMS→RLS scenario, specific parameters are adopted to ensure stability and selective activation: $\lambda_{CAS} = 0.9995$ and $\delta_{CAS} = 100$. Additionally, an error threshold of $\gamma = 0.05$ is established to trigger the RLS update stage.

All variables and vectors used to store weights, errors, outputs, and mixing coefficients are initialized as zero matrices before the adaptation loop commences. In all simulation scenarios, the reference input ($x[n]$) is the noise signal (either white or colored), while the primary/desired signal ($d[n]$) consists of the chirp signal mixed with noise. The complete list of simulation parameters is presented in Table 1.

TABLE 1. Simulation Parameters

Category	Parameter (Symbol)	Value
Signal & Sampling	Sampling Frequency (f_s)	2 MHz
	Signal Duration (T_{obs})	2.5 ms
	Total Samples (N)	$T_{obs} \cdot f_s$
Radar (LFM Pulsed)	Chirp Bandwidth (B)	200 kHz
	Pulse Width (T_p)	1 ms
	Chirp Slope (S)	B/T_p
	Carrier Frequency (f_c)	2.4 GHz
	Target Range (R)	30 km
	Radial Velocity (v)	30 m/s
	Time Delay (τ)	$2R/c$
	Doppler Shift (f_D)	$2v/\lambda$
Radar Performance	Range Resolution (ΔR)	$c/2B$
Adaptive Filter	Filter Order (M)	32
LMS	Step Size (μ)	0.001
	Error Threshold (γ)	0.05
RLS	Forgetting Factor (λ_{RLS})	0.995
	Regularization (δ_{RLS})	1
Cascade	Forgetting Factor (λ_{CAS})	0.9995
	Regularization (δ_{CAS})	100
Noise	White Noise Std Dev (σ)	1.5
	Colored Noise Std Dev (σ)	0.5
	AR(1) coefficient (a)	0.9
Input Signal	Reference signal $x(n)$	–
	Desired signal $d(n)$	–

B. SIMULATION FRAMEWORK AND CONVENTIONAL ALGORITHMS

To evaluate the effectiveness of the noise reduction techniques, this study follows a systematic simulation workflow as illustrated in Fig. 2. The process begins with the initialization of radar and filter parameters, followed by the generation of the LFM chirp signal contaminated by noise (white or colored). The core processing involves an iterative adaptive filtering loop where the input signal $x[n]$ is processed sample by sample to update the filter weights and estimate the clean signal. Finally, the performance is evaluated using Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE) metrics.

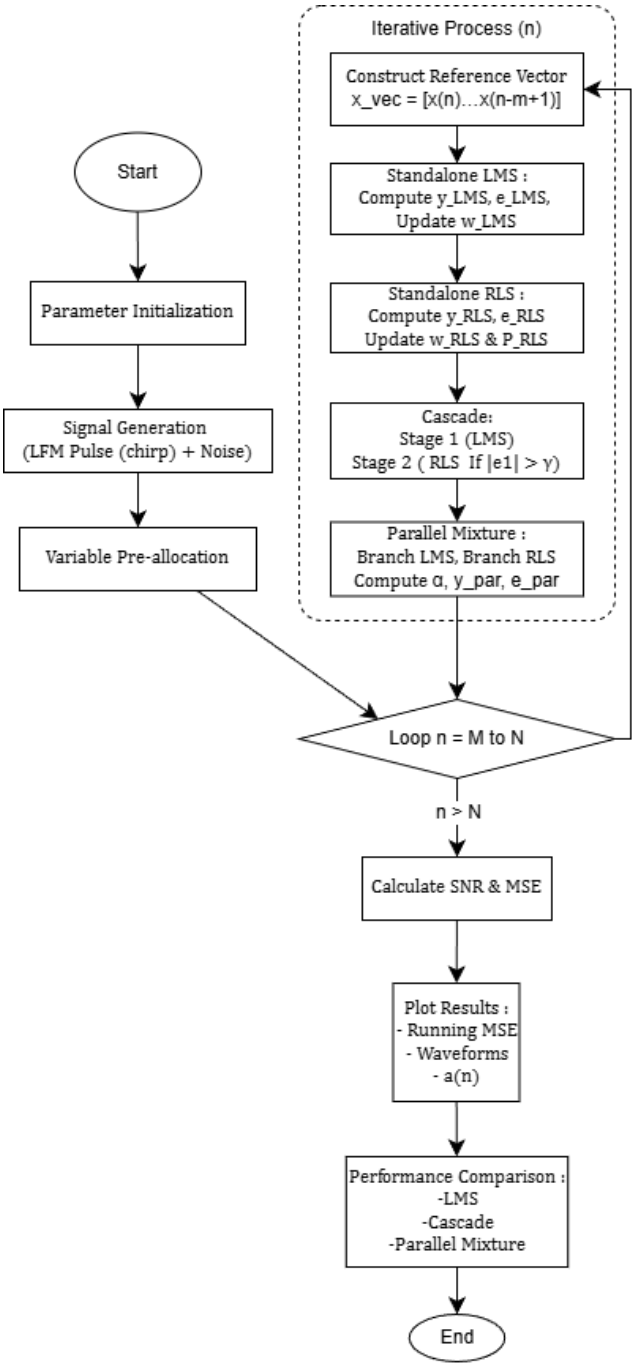


FIGURE 2. Simulation flowchart for the adaptive radar signal processing framework.

Before introducing the proposed hybrid architectures, the baseline performance is established using conventional algorithms: Least Mean Squares (LMS) and Recursive Least Squares (RLS).

1) Least Mean Squares (LMS)

The LMS algorithm is widely adopted due to its computational simplicity, which scales linearly with the filter order $O(M)$. It operates by iteratively adjusting the filter weights to minimize the instantaneous squared error between the desired signal and the filter output.

For each iteration n, the filter output $y(n)$ is computed, and the estimation error $e(n)$ is calculated as the difference between the desired signal $d(n)$ and $y(n)$. The weight update equation is defined as:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu e(n)\boldsymbol{x}^*(n) \tag{13}$$

where $\mathbf{w}(n)$ is the weight vector, $\mathbf{x}(n)$ is the input vector, μ represents the step size (learning rate), and $(\cdot)^*$ denotes the complex conjugate. While robust, LMS often exhibits slow convergence characteristics, particularly when the input signal possesses a large eigenvalue spread, such as in colored noise environments.

2) Recursive Least Squares (RLS)

The RLS algorithm is designed to minimize the weighted cumulative squared error recursively from the initialization to the current time step. This approach allows RLS to achieve significantly faster convergence and better tracking capability in non-stationary environments compared to LMS, albeit at a higher computational cost of $O(M^2)$. The cost function minimized by RLS is given by:

$$J(n) = \sum_{i=1}^{n} \lambda^{n-i} |e(i)|^2 \tag{14}$$

where λ ($0 < \lambda \leq 1$) is the forgetting factor that assigns exponentially lower weights to older data samples.

The update process involves several sequential steps. First, the Kalman gain vector, $k(n)$, is computed using the inverse correlation matrix $\boldsymbol{P}(n-1)$:

$$\mathbf{k}(n) = \frac{\mathbf{P}(n-1)x(n)}{\lambda + \mathbf{x}^H(n)\mathbf{P}(n-1)\mathbf{x}(n)} \tag{15}$$

Next, the *a priori* estimation error is calculated based on the previous weights:

$$e(n) = d(n) - \mathbf{w}^{\boldsymbol{H}}(n-1)\mathbf{x}(n) \tag{16}$$

The filter weights are then updated using the computed gain and error:

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{k}(n)e(n) \tag{17}$$

Finally, the inverse correlation matrix is updated for the next iteration to ensure recursive consistency:

$$\mathbf{P}(n) = \lambda^{-1}\left(\mathbf{P}(n-1) - \mathbf{k}(n)\mathbf{x}^{\boldsymbol{H}}(n)\mathbf{P}(n-1)\right) \tag{18}$$

C. PROPOSED HYBRID ARCHITECTURES

Conventional adaptive filters often face a trade-off between convergence speed and computational complexity. To address these limitations, this study proposes two hybrid architectures that combine the strengths of LMS and RLS algorithms: a selective Cascade scheme and a Parallel Mixture scheme.

1) Cascade LMS-RLS Scheme

The proposed Cascade LMS-RLS architecture aims to balance computational load and estimation accuracy by employing a conditional two-stage filtering process, as illustrated in Fig. 3. This hybrid structure combines the computational simplicity of the Least Mean Squares (LMS) algorithm with the rapid convergence capability of the Recursive Least Squares (RLS) algorithm.

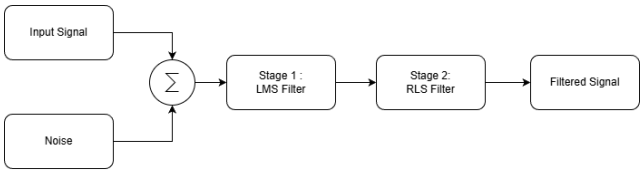


FIGURE 3. Block diagram of the proposed Cascade LMS-RLS architecture with threshold-based activation..

The core innovation of this scheme lies in the selective activation of the second stage using a predefined error threshold, γ_{th} . The operation is divided into two sequential stages:

a: Stage 1: LMS Pre-filtering

The first stage functions as a coarse estimator designed to reduce high-amplitude noise with minimal computational cost. For each time step n, the LMS filter calculates an initial output estimate $y_1^{\text{cas}}(n)$ and the corresponding residual error $e_1^{\text{cas}}(n)$:

$$y_1^{\text{cas}}(n) = (\mathbf{w}_1^{\text{cas}})^T \mathbf{x}_{\text{vec}} \tag{19}$$

$$e_1^{\text{cas}}(n) = d(n) - y_1^{\text{cas}}(n) \tag{20}$$

The weights of the first stage, $\mathbf{w}_1{}^{\text{cas}}$, are continuously updated using the standard gradient descent rule with a fixed step size μ_{LMS}:

$$\mathbf{w}_1^{\text{cas}}(n+1) = \mathbf{w}_1^{\text{cas}}(n) + \mu_{LMS} \boldsymbol{e}_1^{\text{cas}}(n)\mathbf{x}_{\text{vec}} \tag{21}$$

b: Stage 2: Conditional RLS Processing

After the LMS stage, the system evaluates the absolute value of the residual error, $|e_1^{\text{cas}}(n)|$. The logic for the second stage is governed by the threshold parameter γ_{th}:

- **Condition A (Low Error):** If $|e_1^{\text{cas}}(n)| \leq \gamma_{\text{th}}$, the LMS estimation is considered sufficient. The RLS stage is bypassed to conserve resources, and the final error is simply $e_2^{\text{cas}}(n) = e_1^{\text{cas}}(n)$.
- **Condition B (High Error)**: If $|e_1^{\text{cas}}(n)| > \gamma_{\text{th}}$, it indicates significant residual noise or a sudden change in signal characteristics. The RLS filter is activated to refine the estimate.

When activated, the RLS stage estimates the residual component to further minimize the error. The final cascaded error $e_2^{\text{cas}}(n)$ is computed as:

$$e_2^{\text{cas}}(n) = e_1^{\text{cas}}(n) - (\mathbf{w}_2^{\text{cas}})^T \mathbf{x}_{\text{vec}} \tag{22}$$

Subsequently, the RLS weights $\mathbf{w}_2{}^{\text{cas}}$ are updated using the Kalman gain vector $\mathbf{k}_2$ and the inverse covariance matrix $\mathbf{P}_2^{cas}$:

$$\mathbf{k}_2 = \frac{\mathbf{p}_2^{\text{cas}}\mathbf{x}_{\text{vec}}}{\lambda_{\text{CAS}} + \mathbf{x}_{\text{vec}}^{\text{T}}\mathbf{p}_2^{\text{cas}}\mathbf{x}_{\text{vec}}} \tag{23}$$

$$\mathbf{w}_2^{\text{cas}}(n+1) = \mathbf{w}_2^{\text{cas}}(n) + \mathbf{k}_2 e_2^{\text{cas}}(n) \tag{24}$$

where λ_{CAS} is the forgetting factor specific to the cascade stage. This mechanism ensures that the computationally intensive RLS algorithm is only utilized when the LMS filter fails to meet the required accuracy, providing an optimal trade-off between performance and efficiency.

2) Parallel Mixture Scheme

The proposed Parallel Mixture architecture operates by executing two adaptive filter algorithms, LMS and RLS, simultaneously on the same input signal. The core principle of this algorithm is to leverage the distinct advantages of both methods: the computational stability and robustness of the LMS algorithm, and the fast convergence rate of the RLS algorithm.

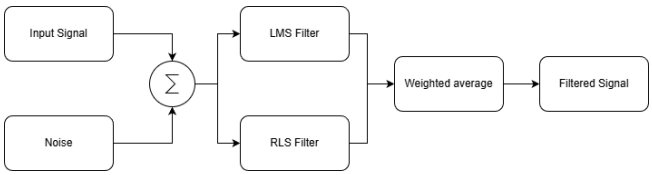


FIGURE 4. Block diagram of the Parallel Mixture LMS+RLS architecture with adaptive mixing coefficient.

Unlike the cascade approach where filters are arranged in series, this scheme processes the input concurrently as illustrated in Fig. 4. The operation is divided into two key components: dual-branch processing and the adaptive mixing mechanism.

a: Dual-Branch Processing

In this scheme, the input vector $\mathbf{x}_{\text{vec}}$ is simultaneously fed into two independent adaptation blocks:

- **LMS Branch:** Computes its output estimate $y_{\text{LMS}}(n)$ and updates its weight vector $\mathbf{w}_{\text{LMS}}$ using the standard stochastic gradient descent method.

- **RLS Branch:** Concurrently computes its output $y_{RLS}(n)$ while updating weights $\mathbf{w}_{RLS}$ and the inverse correlation matrix to minimize the least squares error.

Both filters operate independently in each iteration n, generating their respective instantaneous squared errors, $e^2{}_{LMS}(n)$ and $e^2{}_{RLS}(n)$. These error metrics serve as the control inputs for the mixing mechanism.

b: Adaptive Mixing Mechanism

The critical innovation of this architecture is the dynamic fusion of the two outputs using a time-varying adaptive mixing coefficient, $\alpha(n)$. This coefficient acts as a soft switch based on the relative performance of each filter. It is computed by normalizing the squared errors:

$$\alpha(n) = \frac{e^2_{\mathrm{RLS}}(n)}{e^2_{\mathrm{LMS}}(n) + e^2_{\mathrm{RLS}}(n) + \epsilon} \tag{25}$$

where ϵ is a small regularization constant (10^{-8}) to prevent division by zero.

This formulation provides an intelligent automatic selection mechanism:

- If the RLS filter produces a large error (high $e^2{}_{RLS}$), the numerator increases, causing $\alpha(n)$ to approach 1. Consequently, the system favors the output from the robust LMS branch.
- Conversely, if the RLS filter converges well with low error, $\alpha(n)$ decreases towards 0, shifting the priority to the precise RLS output.

The final combined output, $y_{par}(n)$, is generated through a smooth convex combination defined by:

$$y_{par}(n) = \alpha(n) y_{LMS}(n) + (1 - \alpha(n)) y_{RLS}(n) \tag{26}$$

This weighted averaging strategy ensures that the system continuously balances the trade-off between the low steadystate error of the LMS and the rapid tracking capability of the RLS, providing superior performance in non-stationary noise environments compared to standalone filters.

D. SIMULATION SETUP AND PERFORMANCE METRICS

To validate the effectiveness of the proposed hybrid architectures, simulations were conducted in a discrete-time domain using the parameters previously outlined in Section III-A.

The primary input signal $d(n)$ consists of the LFM chirp target echo mixed with additive noise, while the reference signal $x(n)$ represents the noise source correlated with the interference in the primary channel.

1) Noise Scenarios

Two distinct noise environments are modeled to evaluate the robustness of the algorithms against different spectral characteristics:

- **White Gaussian Noise:** Generated with a zero-mean normal distribution and a standard deviation of $\sigma = 1.5$. This scenario represents uncorrelated wideband interference.
- **Colored Noise:** Modeled using a first-order Autoregressive (AR(1)) process to simulate correlated interference, which is characteristic of radar clutter or jamming. The AR(1) filter is defined with a pole at 0.9 and a standard deviation of $\sigma = 0.5$.

2) Performance Metrics

The filtering performance is quantified using two key metrics: Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE).

The SNR measures the quality of the recovered signal relative to the residual noise power. It is calculated as:

$$\mathrm{SNR} = 10 \log_{10}\left(\frac{\mathrm{var}(s)}{\mathrm{var}(s - \hat{s})}\right) \tag{27}$$

where $\mathrm{var}(s)$ denotes the variance of the clean ground-truth signal $s(n)$, and $e_{out}(n)$ is the denoised output from the adaptive filter (which corresponds to the error signal $e(n)$ in an ANC configuration).

The MSE evaluates the estimation accuracy by measuring the average squared deviation between the recovered signal and the clean reference signal:

$$\mathrm{MSE} = \frac{1}{N}\sum_{n=1}^{N}\big(s(n) - \hat{s}(n)\big)^2 \tag{28}$$

where N is the total number of samples in the observation window.

IV. SIMULATION RESULTS AND ANALYSIS

The simulation parameters, including the radar system specifications (LFM chirp properties) and noise characteristics (white and colored Gaussian noise), have been detailed in Section III. This section presents the performance analysis of the proposed adaptive filters based on the simulation results. The analysis begins with the validation of the radar signal model and continues with a quantitative evaluation using Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE) metrics.

A. RADAR SYSTEM VISUALIZATION AND SIGNAL VALIDATION

Before evaluating the denoising performance, it is essential to validate that the simulated signals accurately represent a monostatic pulsed radar system. Figure 5 illustrates the real part of the Transmit (Tx) signal compared to the received Echo signal. The plot demonstrates that the echo is not merely a continuous signal but a reflected pulse that appears after a specific time delay.

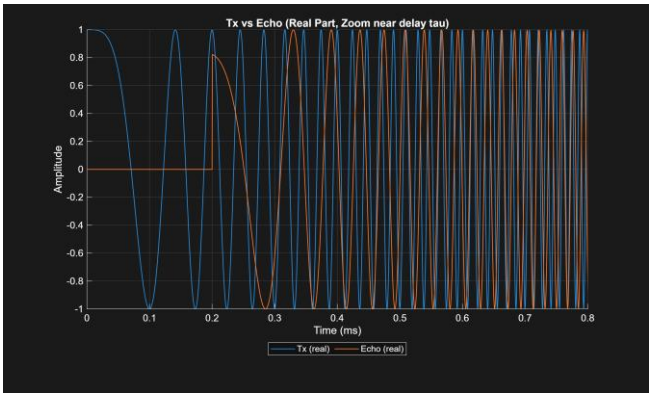


FIGURE 5. Comparison of the Transmit (Tx) signal and the received Echo signal (Real Part), showing the pulse structure.

To observe the time delay more clearly, the signal envelopes are presented in Fig. 6. The time delay (τ) observed between the start of the Tx pulse and the Echo pulse corresponds to the round-trip propagation time for a target located at a distance $R = 30$ km.

Furthermore, although the envelope plot primarily highlights the range delay, the underlying carrier of the echo signal incorporates the Doppler shift effect (f_D). This shift is induced by the target's radial velocity of 30 m/s relative to the radar. While the Doppler effect is subtle in the time domain envelope, it is mathematically embedded in the phase of the received signal, confirming that the simulation model validly represents a moving target scenario rather than a static reflection.

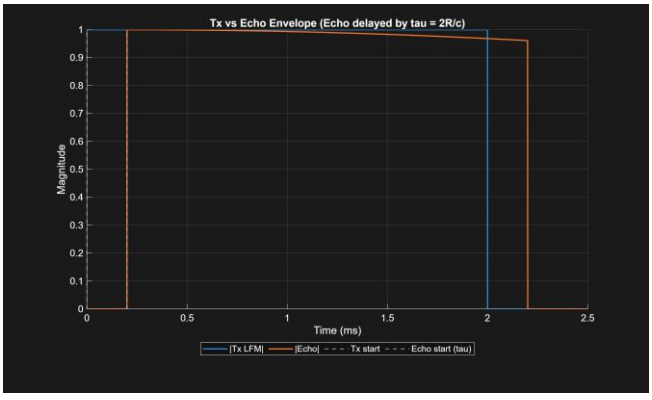


FIGURE 6. Envelope comparison between Tx and Echo signals. The observed delay τ validates the target range parameter (R).

The ultimate validation of the radar signal processing chain is demonstrated through the Pulse Compression (Matched Filter) output, as shown in Fig. 7. This range profile represents the radar's ability to detect the target.

In the raw noisy signal (indicated by the blue line), the target peak is obscured by a high noise floor, making reliable detection difficult. However, after applying the proposed adaptive filtering methods—specifically the Parallel Mixture and Cascade LMS-RLS—the noise floor is significantly suppressed. This suppression reveals a distinct mainlobe peak at the correct range bin. The preservation of the peak location confirms that the denoising process improves the Signal-to-Noise Ratio (SNR) without introducing timing distortions, thereby enhancing the robustness of target detection in noisy environments.

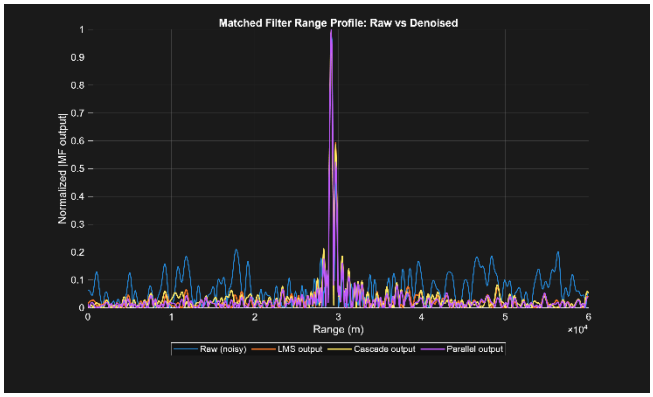


FIGURE 7. Matched Filter Range Profile comparing the raw noisy signal (blue) against the denoised outputs. The adaptive filters significantly lower the noise floor, making the target peak distinguishable.

B. PERFORMANCE ANALYSIS IN WHITE GAUSSIAN NOISE SCENARIO

In this scenario, the radar signal is corrupted by white Gaussian noise (σ = 1.5), resulting in a low baseline SNR of −6.70 dB. The performance of the three adaptive filtering approaches—Standalone LMS, Cascade LMS-RLS, and Parallel Mixture—is evaluated based on their ability to recover the chirp waveform and improve the SNR.

1) Waveform Reconstruction Analysis

The visual comparison of the denoised signals reveals distinct differences in the capabilities of each algorithm.

Figure 8 shows the output of the Standalone LMS filter. While the algorithm manages to reduce some of the high frequency noise, the output (blue line) still exhibits significant residual fluctuations, particularly during the initial adaptation phase. This aligns with the known limitation of LMS, which suffers from slow convergence when the step size is small.

In contrast, the Cascade LMS-RLS method (Figure 9) demonstrates improved stability. By activating the RLS stage only when the LMS error exceeds a threshold (γth = 0.05),

this method effectively corrects large deviations without the computational burden of a fully continuous RLS. The resulting waveform follows the linear chirp pattern more closely than the LMS output.

However, the most superior reconstruction is achieved by the Parallel Mixture method, as shown in Fig. 10. The output signal is remarkably smooth and aligns almost perfectly with the clean reference signal. The adaptive mixing mechanism allows the system to seamlessly switch between LMS and RLS outputs, effectively suppressing noise artifacts that persist in the other methods.

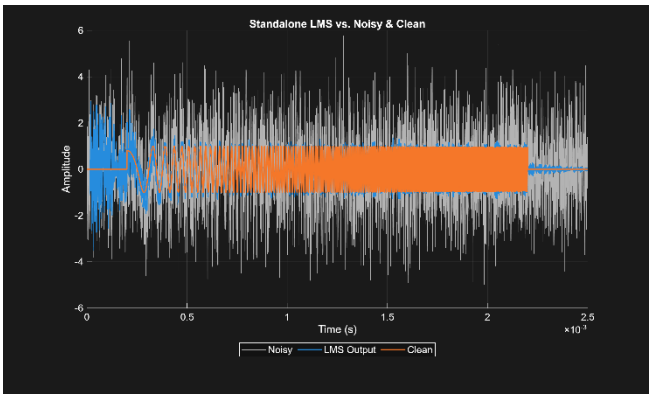


FIGURE 8. LMS filter output under White Gaussian Noise. Residual noise remains visible

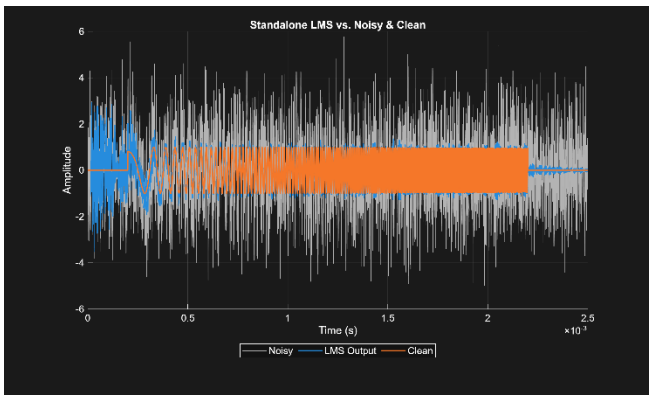


FIGURE 9. Cascade LMS-RLS output showing improved stability compared to standalone LMS.

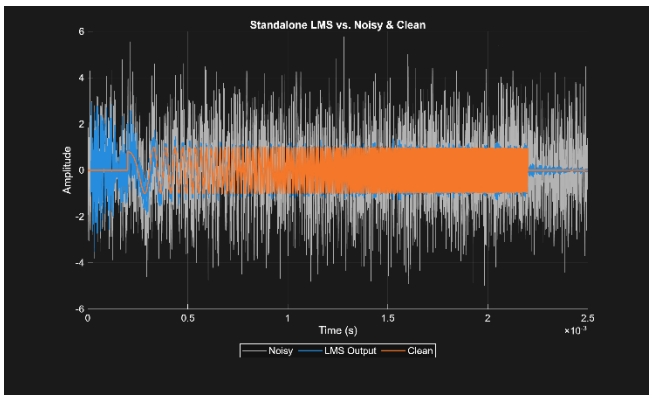


FIGURE 10. Parallel Mixture output under White Gaussian Noise. The waveform is smooth and closely matches the clean signal.

2) Convergence and Error Analysis

The convergence speed and stability of the filters are analyzed using the Running Mean Squared Error (MSE) metric, presented in Fig. 11.

The Standalone LMS (blue line) exhibits high initial error spikes and a slow decay rate, confirming its sluggish adaptation. The Cascade method (orange line) improves upon this with faster convergence. Notably, the Parallel Mixture (yellow line) maintains the lowest error consistently throughout the simulation duration. It achieves rapid stability almost immediately after the signal onset, demonstrating its robustness against random white noise fluctuations.

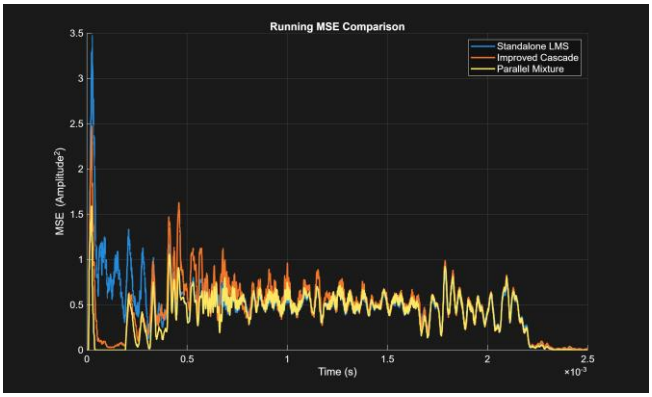


FIGURE 11. Comparison of Running MSE for LMS, Cascade, and Parallel methods in White Noise. The Parallel method (yellow) shows the fastest convergence and lowest steady-state error.

3) Quantitative Performance (SNR and MSE)

Table 2 summarizes the quantitative improvements. The Standalone LMS provides a modest improvement, raising the SNR from -6.70 dB to -0.38 dB. The Cascade approach offers a better enhancement, achieving an SNR of 1.68 dB.

The Parallel Mixture approach outperforms both, achieving a significant SNR increase to 4.02 dB and reducing the MSE to 0.48895. This substantial gain (> 10 dB improvement from baseline) confirms that dynamically combining the stability of LMS with the precision of RLS yields the optimal solution for white noise environments.

TABLE 2. Performance Comparison in White Gaussian Noise

Method	SNR (dB)	MSE
Before Filtering (Baseline)	-6.70	2.68723
Standalone LMS	-0.38	1.01852
Cascade LMS → RLS	1.68	0.73961
Parallel Mixture (Proposed)	4.02	0.48895

C. PERFORMANCE ANALYSIS IN COLORED NOISE SCENARIO

This section evaluates the system's performance under Colored Gaussian Noise conditions. Unlike white noise, colored noise possesses temporal correlation and uneven power spectral density, presenting a significantly more challenging environment for adaptive filters. The baseline SNR in this scenario drops to a critical level of -10.31 dB.

1) Waveform Reconstruction Analysis

Visual comparison of the filtered outputs demonstrates how each method handles correlated interference.

1) Standalone LMS Failure: As shown in Fig. 12, the LMS algorithm struggles significantly. Due to its assumption of uncorrelated input, LMS fails to distinguish the correlated noise from the signal features. The output (blue line) remains noisy with severe amplitude distortions, resulting in a negligible SNR improvement of only 0.07 dB.
2) Cascade LMS-RLS Improvement: The Cascade method (Figure 13) offers a noticeable improvement over LMS. By utilizing the RLS stage when the error is high, the system manages to correct some of the distortions caused by the colored noise. The waveform structure is more coherent than the LMS output, achieving an SNR of 1.08 dB. However, some noise artifacts persist because the RLS is not continuously active.
3) Parallel Mixture Superiority: The Parallel Mixture method (Figure 14) yields the most robust reconstruction. It successfully suppresses the correlated noise, producing a smooth chirp waveform that aligns closely with the clean reference. With a peak SNR of 5.18 dB, it outperforms the Cascade method by a wide margin, proving that simultaneous processing is more effective for complex noise than sequential processing.

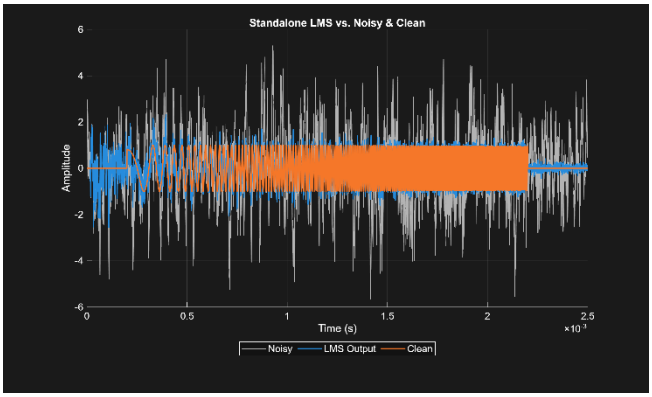


FIGURE 12. LMS output in Colored Noise. The filter fails to recover the signal structure.

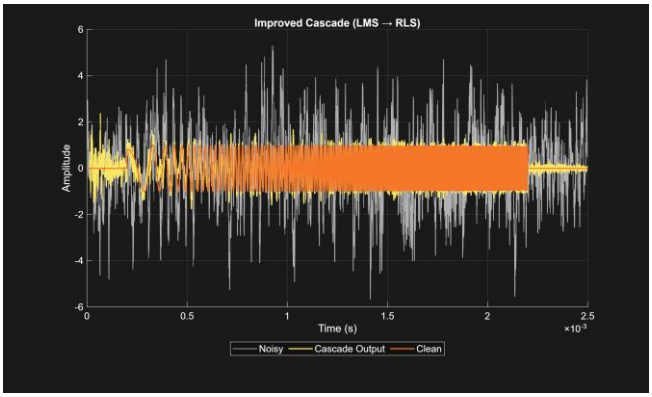


FIGURE 13. Cascade output in Colored Noise. Shows better reconstruction than LMS but still contains artifacts.

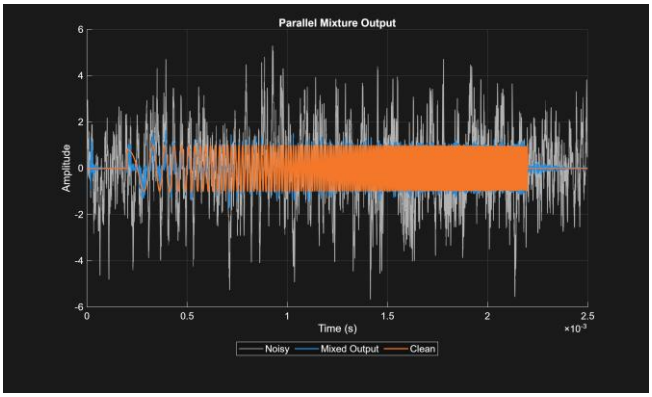


FIGURE 14. Parallel Mixture output in Colored Noise. The waveform is remarkably smooth and accurate.

2) Convergence and Error Analysis (Running MSE)

The stability of the filters in colored noise is analyzed using the Running MSE metric, as shown in Fig. 15.

The LMS algorithm (blue line) exhibits a very high and fluctuating error profile throughout the simulation, confirming its inability to converge effectively in this environment. The Cascade method (orange line) reduces the average error but still shows intermittent spikes corresponding to the switching moments.

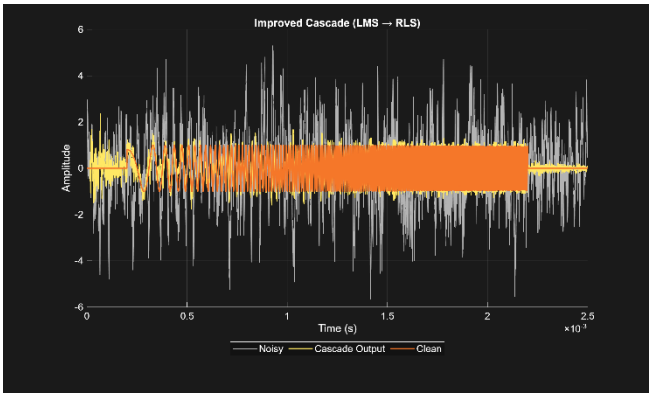


FIGURE 15. Running MSE comparison in Colored Noise. Parallel Mixture (yellow) consistently achieves the lowest error.

3) Quantitative Summary

Table 3 summarizes the performance metrics. The data highlights the significant gap between the proposed Parallel method and the others. While LMS offers almost no improvement (< 1 dB gain) and Cascade provides moderate gain, the Parallel Mixture provides a substantial gain of over 15 dB relative to the baseline, reducing the MSE to 0.45637.

TABLE 3. Performance Comparison in Colored Noise

Method	SNR (dB)	MSE
Before Filtering (Baseline)	-10.31	5.32817
Standalone LMS	0.07	1.07686
Cascade LMS → RLS	1.08	0.86476
Parallel Mixture (Proposed)	5.18	0.45637

D. ADAPTIVE MIXING ANALYSIS

The core innovation of the proposed Parallel Mixture method lies in its ability to dynamically weight the contributions of the LMS and RLS filters. This is governed by the adaptive mixing coefficient, $\alpha(n)$, which is calculated based on the instantaneous squared errors of both filters. A value of $\alpha(n)$ close to 1 indicates a preference for the LMS output, while a value close to 0 favors the RLS output.

This section analyzes the behavior of $\alpha(n)$ under both noise scenarios to understand how the system adapts to different interference characteristics.

1) Behavior in White Gaussian Noise

Figure 16 illustrates the evolution of the mixing coefficient over time when the system is subjected to white Gaussian noise.

As observed, the coefficient fluctuates rapidly but maintains a distribution where higher values (closer to 1) are frequently visited. This behavior suggests that in an uncorrelated noise environment, the LMS filter remains a competent estimator. The system does not strictly discard the LMS output; instead, it leverages the LMS stability while using the RLS component to correct transient errors. The relatively high presence of LMS contribution aligns with the theoretical expectation that LMS performs adequately when the noise spectral density is flat.

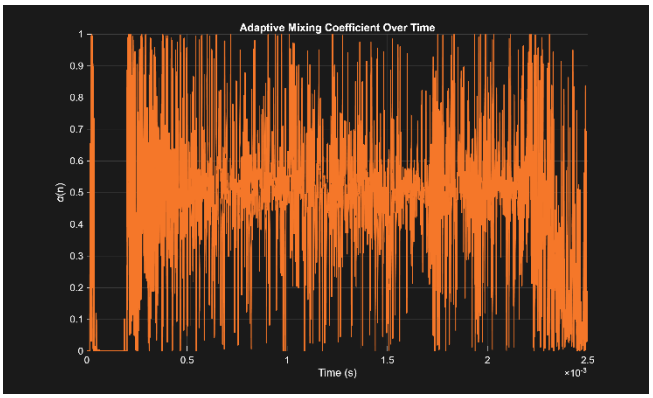


FIGURE 16. Evolution of the adaptive mixing coefficient $\alpha(n)$ in White Gaussian Noise. The system frequently utilizes the LMS output (higher α) due to the uncorrelated nature of the noise.

2) Behavior in Colored Noise

In the colored noise scenario, the behavior of $\alpha(n)$ changes significantly, as shown in Fig. 17.

Unlike the white noise case, the coefficient exhibits extreme volatility, spanning the entire range from 0 to 1 with a more uniform distribution. The frequent dips toward 0 indicate that the system heavily relies on the RLS filter to handle the temporal correlation of the noise—a task where LMS fails.

This dynamic switching capability is the key reason for the superior SNR performance observed in Section IV-C. By continuously evaluating the instantaneous performance of both filters, the Parallel Mixture method effectively "selects" the best filter for every sample. This adaptability allows the system to remain robust even when the noise characteristics are complex and time-varying, proving that the hybrid architecture offers a level of intelligence not present in fixed or threshold-based (Cascade) systems.

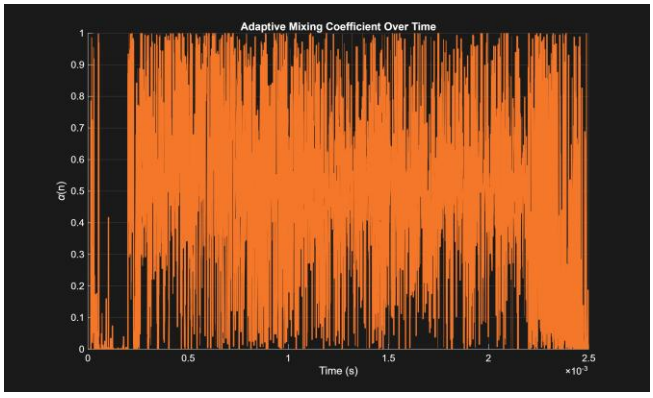


FIGURE 17. Evolution of $\alpha(n)$ in Colored Noise. The coefficient fluctuates intensely across the full range, demonstrating the system's active balancing of LMS and RLS to mitigate correlated interference.

V. CONCLUSION

Based on the simulation results and performance analysis of adaptive filter algorithms on LFM pulsed radar systems, this study concludes the following:

1) The LMS (Least Mean Squares) algorithm proves effective in handling White Gaussian Noise with moderate SNR improvement, but its performance declines drastically in Colored Noise. This is due to the LMS's inability to handle temporal correlation in colored noise, indicated by minimal SNR improvement in that scenario.
2) The hybrid Cascade LMS-RLS method successfully overcomes the limitations of the standalone LMS through a selective correction mechanism based on an error threshold. By activating the RLS only during critical conditions, this method provides more accurate and efficient signal estimation, particularly in Colored Noise scenarios, compared to using LMS alone.
3) The Parallel Mixture method proves to be the most robust approach with the best consistent performance across all test scenarios. This method yields the highest SNR (reaching 5.18 dB in Colored Noise) and the lowest MSE compared to other methods, confirming that simultaneous output combination is the most optimal strategy for complex interference environments.
4) Analysis of the adaptive mixing coefficient $\alpha(n)$ demonstrates that the Parallel Mixture system possesses dynamic real-time adaptation capabilities to environmental characteristics. The system automatically balances the contribution between LMS and RLS according to the encountered noise type (LMS dominance in White Noise and balanced in Colored Noise), without requiring prior knowledge of interference statistics.
5) The application of hybrid adaptive filters, particularly the Parallel Mixture, has a significant impact on the quality of the radar range profile after pulse compression. Filtering results show a tangible reduction in the noise floor around the target, making the range peak more distinct and clearly separated from background interference, which practically enhances target detection clarity.

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